# Internship Progress Report

CamGeo Project — Preparation, Photo-interpretation, and Data Quality

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|---|---|
| **Host organization:** | Novalitix |
| **Intern:** | Fofack Henri |
| **Project:** | CamGeo land cover mapping in Cameroon |
| **Target milestone:** | Gate G2 — GitHub issue #2 — release v0.3.0 |
| **Pilot area:** | Southern Region (cocoa basin) |
| **Period:** | *October 5, 2026 – October 12, 2026* |
| **Supervisors:** | *BRELL SANWOUO & LEONEL* |

## 1. Objectives Recap

The internship aims to build, check, and document the reference sample dataset needed to train the land cover mapping models for the CamGeo project. The main deliverable is a labeled and reviewed sample dataset of at least 350 points over the Southern Region, compliant with the `samples/README.md` schema, and directly usable by scikit-learn and the Google Earth Engine (GEE) pipeline.

This work corresponds to the **Gate G2** milestone (GitHub issue #2, release v0.3.0). It conditions the next stages of the project, led by interns 2 and 3 (classifier training and accuracy assessment).

## 2. Work Completed

### 2.1 GEE Environment Setup

Registration and validation of the Google Earth Engine account. Creation of the associated Google Cloud project and initial hands-on work in the Code Editor.

### 2.2 Sample Point Generation

Execution of the `gee/sampling/generate_sample_points.js` script over the Southern Region. The script generated a reproducible random draw (fixed seed) of 349 points in the pilot area.

### 2.3 Data Export and Preparation

Export of the CSV file from GEE to Google Drive, followed by processing with a Python script in Google Colab:

- ☞ **renaming and normalizing columns;**
- ☞ **adding the 14 columns of the CamGeo schema;**
- ☞ **verifying WGS84 coordinates (EPSG:4326), identifier uniqueness, absence of duplicates, and missing values;**
- ☞ **producing two files:**
  - ☞ `samples_sud_v01_ceo.csv` — for import into CEO;
  - ☞ `samples_sud_v01.csv` — final CamGeo deliverable.

### 2.4 Collect Earth Online Project Setup

Creation of the `CamGeo - Sud - v01` project with:

- ☞ **four collection questions:** `class_code`, `class_name`, `confidence`, `notes`;
- ☞ **imagery sources:** Mapbox Satellite, Sentinel-2, Landsat 8 & 9, Dynamic World, OpenStreetMap;
- ☞ **100 m square plot (1 ha), one sample per plot;**
- ☞ **double review and quality audit planned in the protocol.**

The figure below shows the points generated in the Southern Region.

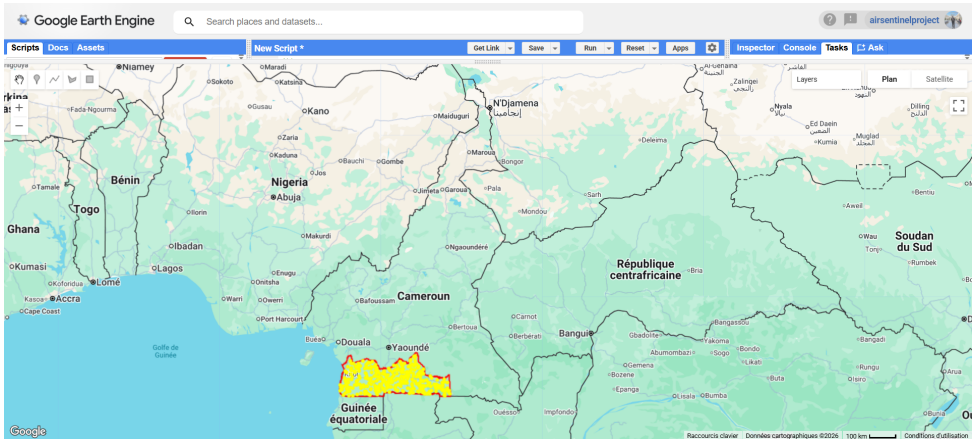



Figure 1: The 349 generated points over the Southern Region.

### 2.5 Completion of the Annotation Campaign

The annotation campaign was completed between **October 5 and October 12, 2026**. All **349 points have been labeled** using the 10-class CamGeo legend, on the basis of Mapbox Satellite, Sentinel-2, Landsat 8 & 9 and Dynamic World imagery. Each point was assigned a dominant class, a confidence level (`high`, `medium`, `low`), and, where relevant, a short note documenting ambiguous cases (cloud cover, class boundaries, proximity to villages or paths). A first export of the annotated data was performed from CEO on **October 12, 2026**, in both `sample-data` and `plot-data` formats.

## 3. Progress Status

Overall progress on the annotation phase is now **100%**. The annotation campaign is closed; the next stage focuses on quality control (double review, audit) and predictor extraction.

## 4. Challenges Encountered and Proposed Solutions

- ☞ **Distinguishing dense forest (class 1) from shaded agroforestry (class 3).** These two classes have similar visual signatures and accounted for most of the ambiguity across the 349 points. Solution: reliance on landscape context (proximity to villages and roads), the presence of emergent trees, and canopy texture, as recommended in the annotation guide.

| Step | Done | Total |
|---|---|---|
| Points generated (GEE) | 349 | 349 |
| Points annotated (CEO) | 349 | 349 |
| Points reviewed (double review) | 0 | 349 |
| Random quality audit (10%) | 0 | 35 |
| GEE predictor extraction | not started | |
| Delivery of `samples_sud_v01.csv` | in progress | |

Table 1: Progress dashboard

| Code | Class name | Count |
|---|---|---|
| 1 | dense humid forest | ∼100 |
| 2 | degraded secondary forest | ∼60 |
| 3 | shaded agroforestry | ∼130 |
| 4 | smallholder agriculture | ∼35 |
| 5 | industrial plantation | ∼20 |
| 6 | mangrove | 1 |
| 7 | savanna grassland | 0 |
| 8 | water | 1 |
| 9 | urban built-up | 4 |
| 10 | bare soil / mining | 0 |

Table 2: Class distribution in the 349 annotated points (indicative)

☞ **Ambiguous cases and cloudy imagery.** Several points were affected by haze, cloud cover, or lay on class boundaries. Solution: assignment of the `low` confidence status and systematic documentation in the `notes` field, to be re-examined during the double review phase.

☞ **Getting started with the CEO interface.** Initial setup took several hours before project creation worked (six unsuccessful attempts). Once configured, the interface proved stable for the rest of the campaign.

☞ **Class imbalance in the Southern cocoa basin.** The annotated sample is dominated by shaded agroforestry (class 3) and dense humid forest (class 1), while mangrove, water, savanna and bare soil (classes 6, 7, 8, 10) are very rare or absent. This imbalance must be documented and taken into account in the next stages.

## 5. Next Steps

1. Set up the systematic double review of the 349 annotated points (`review_status: reviewed`).
2. Perform the random 10% quality audit (approximately 35 points) and compute the inter-annotator agreement rate.
3. Reconstruct the final `samples_sud_v01.csv` file from the CEO export, ensuring strict compliance with the 14-column CamGeo schema.
4. Extract geospatial predictors via GEE: Sentinel-2 bands, NDVI, NDWI, NDBI, Sentinel-1 SAR backscatter ratio (VV, VH, VV/VH), SRTM elevation, slope, aspect, and CHIRPS rainfall accumulations.
5. Document class counts and analyze imbalances specific to the Southern cocoa basin.

6. Deliver `samples_sud_v01.csv`, the GeoParquet version, the quality control report, and the methodological note, for the closure of the Gate G2 milestone.

## 6. Conclusion

The annotation campaign has been completed successfully: all **349 points over the Southern Region are now labeled** in Collect Earth Online, marking the end of the data collection phase. The next priority is to secure annotation quality through systematic double review and the 10% random audit, then to reconstruct the final `samples_sud_v01.csv` file and extract the geospatial predictors, in order to deliver a sample dataset fully compliant with the Gate G2 milestone.